

Splitting a Concrete Defect Extension in a Restricted Mark Cokernel

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Abstract

For a frozen 16×16 restricted self-mark matrix M , earlier exact calculations locate a subgroup $D \cong \mathbb{Z}/8 \oplus (\mathbb{Z}/2)^2$ inside $C = \mathbb{Z}^{16}/M\mathbb{Z}^{16}$. Abstract invariant factors do not decide whether this particular inclusion splits. We give a three-coordinate retraction, an explicit index-32 lattice for its kernel, and an integral presentation of the resulting complement. We also prove that the fixed subgroup has exactly 2^{41} complements. All claims concern only the named sixteen-type support.

1 The concrete extension

Let M be the exact restricted mark matrix, let $U = [u_1 \ u_2 \ u_3]$, and put

$$C = \mathbb{Z}^{16}/M\mathbb{Z}^{16}, \quad A = \mathbb{Z}/8 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

The preceding certificate proves that $a \mapsto [Ua]$ embeds A in C ; write its image as D . It also gives

$$C/D \cong (\mathbb{Z}/2)^8 \oplus (\mathbb{Z}/4)^2 \oplus \mathbb{Z}/12 \oplus \mathbb{Z}/144.$$

The question here is about the actual extension $0 \rightarrow D \rightarrow C \rightarrow C/D \rightarrow 0$. The abstract groups alone are insufficient: $0 \rightarrow \mathbb{Z}/2 \rightarrow \mathbb{Z}/4 \rightarrow \mathbb{Z}/2 \rightarrow 0$, for example, does not split.

The scope firewall is

`NO_BAD_EULER_OR_ROOT_NUMBER.`

No full table of marks, full Burnside ring, arithmetic or local construction, Euler factor, root number, automorphy, or Hilbert–Polya operator is claimed.

2 A sparse retraction

For $x = (x_1, \dots, x_{16})^T$, define

$$\rho([x]) = (x_{10} \bmod 8, \ x_3 \bmod 2, \ x_1 + x_{15} \bmod 2).$$

Its integer row matrix is

$$R = \begin{pmatrix} e_{10}^T \\ e_3^T \\ e_1^T + e_{15}^T \end{pmatrix}, \quad \Delta = \text{diag}(8, 2, 2).$$

Theorem 1 (Splitting of the actual inclusion). *The map $\rho : C \rightarrow A$ is well defined and its restriction to D is the identity. Consequently $C = D \oplus \ker \rho$.*

Proof. Exact integer multiplication gives $RM \equiv 0$ rowwise modulo $(8, 2, 2)$, so ρ descends through $M\mathbb{Z}^{16}$. It also gives

$$RU = \text{diag}(1, -1, -1) \equiv I_3 \pmod{\Delta}.$$

Thus $\rho([Ua]) = a$ in A for every a , which is precisely the retraction identity for the fixed inclusion. $\square$

3 An explicit complement

Let

$$L = \{x \in \mathbb{Z}^{16} : x_{10} \equiv 0 \pmod{8}, x_3 \equiv 0 \pmod{2}, x_1 + x_{15} \equiv 0 \pmod{2}\}.$$

This is the full inverse image of zero under the congruence map represented by R ; in particular, it is not the integer kernel of R . A column basis B is obtained from the standard basis by replacing

$$b_1 = e_1 - e_{15}, \quad b_3 = 2e_3, \quad b_{10} = 8e_{10}, \quad b_{15} = 2e_{15}.$$

It has determinant 32. Since $RM \equiv 0 \pmod{\Delta}$, $M\mathbb{Z}^{16} \subset L$.

Proposition 1 (Complement presentation). *Put $K = L/M\mathbb{Z}^{16}$. The matrix $N = B^{-1}M$ is integral, satisfies $BN = M$, and has Smith invariants*

$$(1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 4, 4, 12, 144).$$

Hence

$$K \cong (\mathbb{Z}/2)^8 \oplus (\mathbb{Z}/4)^2 \oplus \mathbb{Z}/12 \oplus \mathbb{Z}/144, \quad |K| = 7077888.$$

Proof. The displayed basis describes the three congruences directly. Exact Gauss–Jordan elimination makes N integral; Euclidean Smith reduction gives the stated diagonal. The product check $|D||K| = 32 \cdot 7077888 = 226492416 = |C|$ is independent confirmation. $\square$

4 Counting every complement

Fix the displayed retraction ρ_0 . Any other retraction differs from it by a homomorphism vanishing on D , hence by a unique member of $\text{Hom}(C/D, D)$. Conversely every such homomorphism, composed with the quotient map and added to ρ_0 , is a retraction. Thus the retractions form a torsor under $\text{Hom}(C/D, D)$.

The 2-primary part of the quotient is

$$(C/D)_{(2)} \cong (\mathbb{Z}/2)^8 \oplus (\mathbb{Z}/4)^3 \oplus \mathbb{Z}/16.$$

Using $|\text{Hom}(\mathbb{Z}/2^a, \mathbb{Z}/2^b)| = 2^{\min(a,b)}$, the exponents for the three target factors $\mathbb{Z}/8, \mathbb{Z}/2, \mathbb{Z}/2$ are respectively 17, 12, 12. Therefore

$$|\text{Hom}(C/D, D)| = 2^{17+12+12} = 2^{41} = 219902325552.$$

For a fixed inclusion, taking kernels bijects retractions with complements: a chosen complement determines the unique projection that is the identity on D . Hence there are exactly 2^{41} complements. In particular, the explicit K above is not canonical.

5 Reproducibility and conclusion

The evidence binds the exact C64, C65, C66, and C68 bytes. Producer and checker independently reconstruct R, B, N , the Smith diagonal, and the homomorphism count; a separate SymPy path reproduces the matrix identities and Smith form. Clean replay passes and twenty-three hostile semantic mutations are rejected. C69 therefore resolves the extension question constructively: the concrete defect subgroup splits, its displayed complement has the C68 quotient invariants, and all complements are counted exactly.